

WaRee11

iPadOS River Simulation & Hydraulic Modeling Platform

Public White Paper Brief v0.1



For partners, investors, university laboratories, academic collaborators, and technical advisors

Status: Concept / Prototype Direction | Date: May 2026

1. Executive Brief

WaRee11 is an iPadOS-based river simulation and hydraulic modeling platform designed to make river model preparation, hydraulic education, scenario exploration, and results communication more accessible through a touch-first iPad workflow.

The platform is inspired by professional hydraulic modeling systems such as HEC-RAS, MIKE 11 / MIKE+ Rivers, SOBEK, and D-Flow FM, but it is not positioned as a direct replacement at this stage. Its early role is best understood as a focused companion platform for learning, field preparation, model readiness checking, and structured scenario thinking.

WaRee11's initial direction emphasizes a one-dimensional river modeling workflow, cross-section management, simulation settings, CFL-aware diagnostics, scenario comparison, and future GIS-assisted model preparation. The long-term ambition is to connect classroom learning, field observation, and professional river modeling practice.

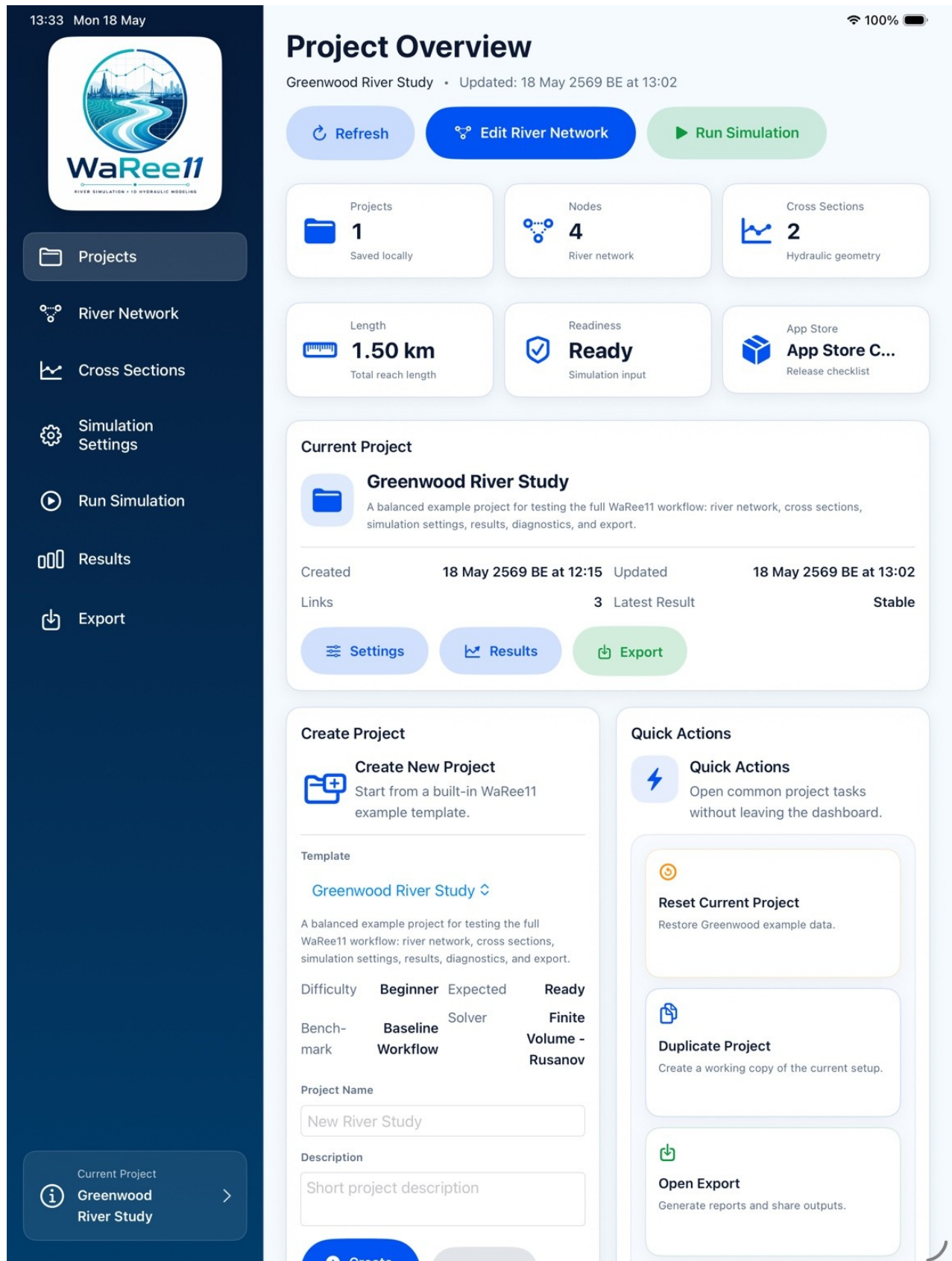


Figure 1. WaRee11 project dashboard showing project readiness and workflow navigation.

2. Why WaRee11 Matters

Value Area	Contribution
Education	Provides students and instructors with a visual iPad environment for understanding river networks, cross-section geometry, Manning roughness, boundary conditions, CFL stability, and simulation results.
Field preparation	Helps engineers and field teams review river networks, cross-section completeness, roughness assumptions, and model readiness away from a desktop workstation.
Scenario exploration	Supports rapid what-if comparison of assumptions such as roughness, hydrographs, downstream water levels, and simplified design configurations.

3. Positioning

WaRee11 should be presented as a touch-first iPadOS companion platform, not as a full replacement for mature professional desktop modeling systems. This positioning keeps the project credible while highlighting its unique role in education, field review, and early scenario analysis.

Classroom Hydraulics → WaRee11 → Professional Desktop Modeling

Supported by field GIS / iPad-based model preparation workflows

4. Core WaRee11 Modules

Module	Role in the Workflow
Project Dashboard	Project health, readiness status, recent runs, and export shortcuts.
River Network Editor	River nodes, branches, chainage, connectivity, and topology checks.
Cross-Section Editor	Geometry input, Manning roughness, cross-section preview, hydraulic properties, and rating curve preview.
Simulation Settings	Time step, duration, CFL checking, numerical scheme selection, and pre-run diagnostics.
Results & Diagnostics	Water level charts, discharge charts, run summaries, solver warnings, result tables, and export-ready outputs.

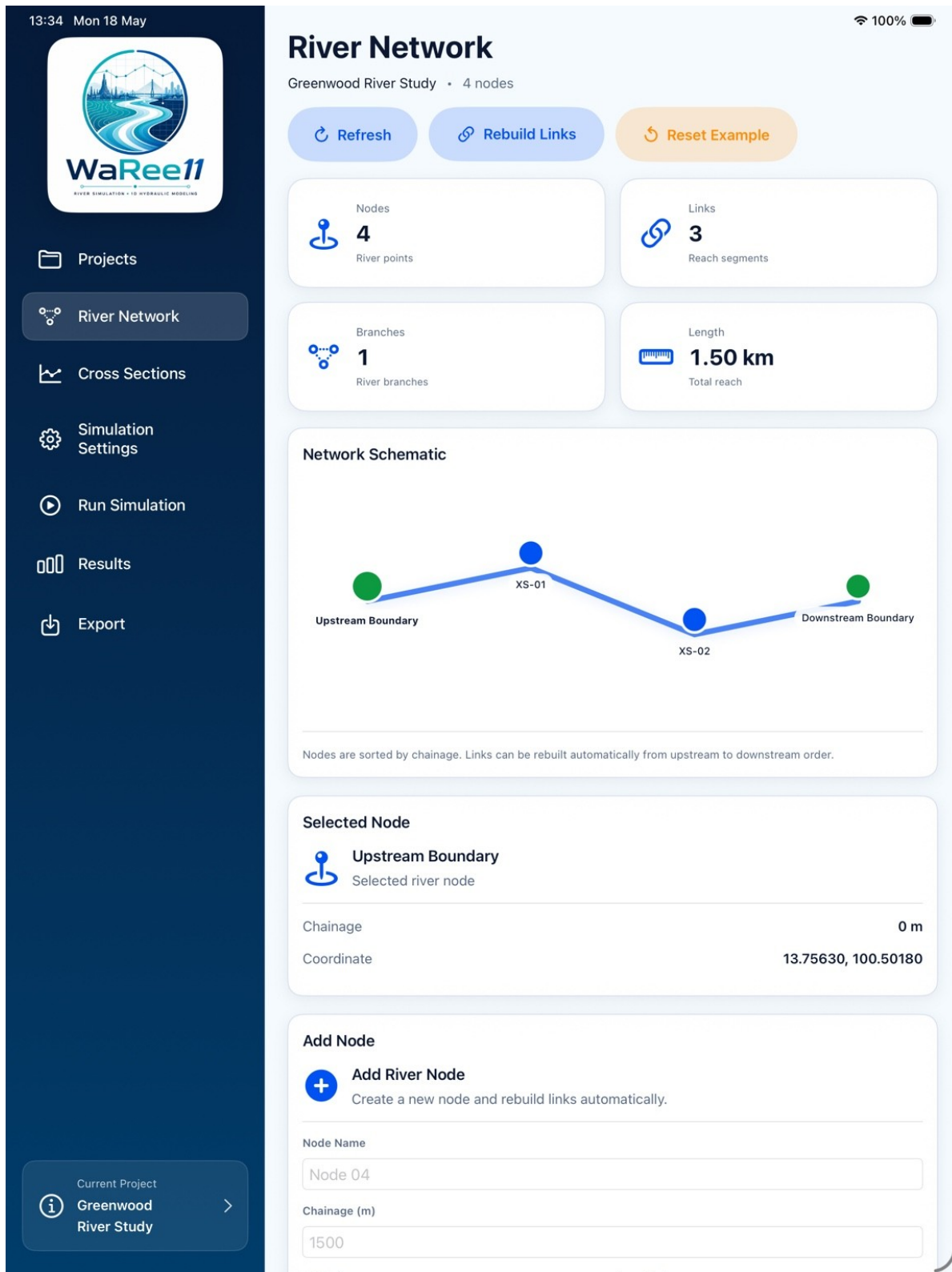


Figure 2. River network editor showing nodes, links, branches, and reach length.

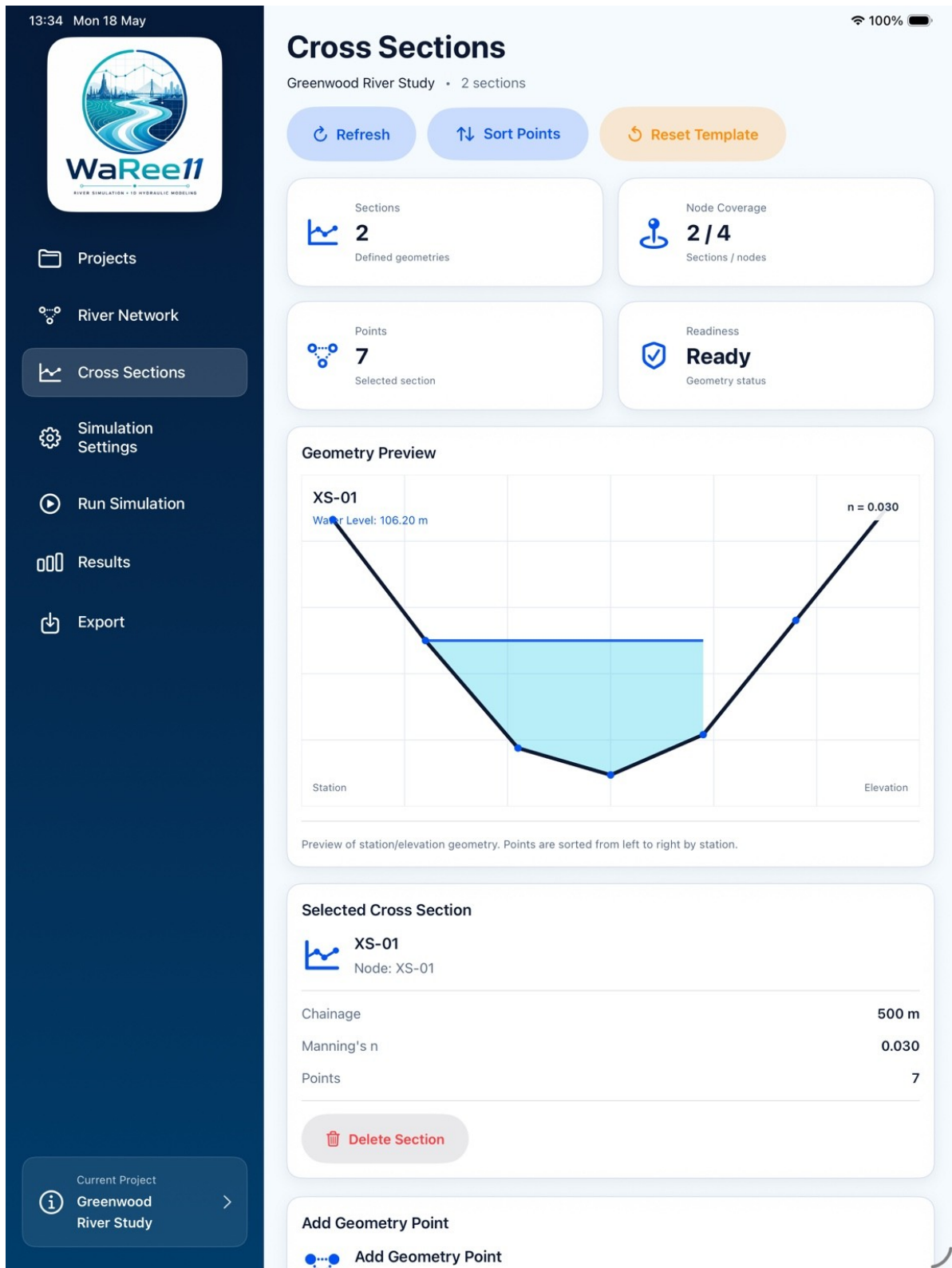


Figure 3. Cross-section editor showing geometry preview and hydraulic section data.

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Projects

River Network

Cross Sections

Simulation Settings

Run Simulation

Results

Export

Current Project

Greenwood River Study

Simulation Settings

Greenwood River Study • Δt 1 s • Duration 3600 s • CFL 0.850

Save Settings

Reset Defaults

Apply Recommended Δt

Time Step

1 s

Numerical Δt

Duration

1.00 hr

Total run time

CFL

0.850

User target

Validation

Ready

Settings status

Validation

Ready

Settings are valid for a baseline prototype run.

Blocking

0

Warnings

0

Info

1

Settings Ready

Simulation settings are valid for a baseline prototype run.

Recommendation: You can proceed to Run Simulation.

Auto-save

Auto-save Ready

Changes will be saved automatically.

Simulation settings are automatically saved after editing pauses. Leaving this page also flushes pending changes.

Time Control

Time Step

Calculation interval in seconds

1 s

Total Duration

Total simulation length

3600 s

Courant Number

Recommended range: 0.3–1.0

0.850 -

Duration Display

1.00 hr

Figure 4. Simulation settings with time step, duration, CFL target, and readiness validation.

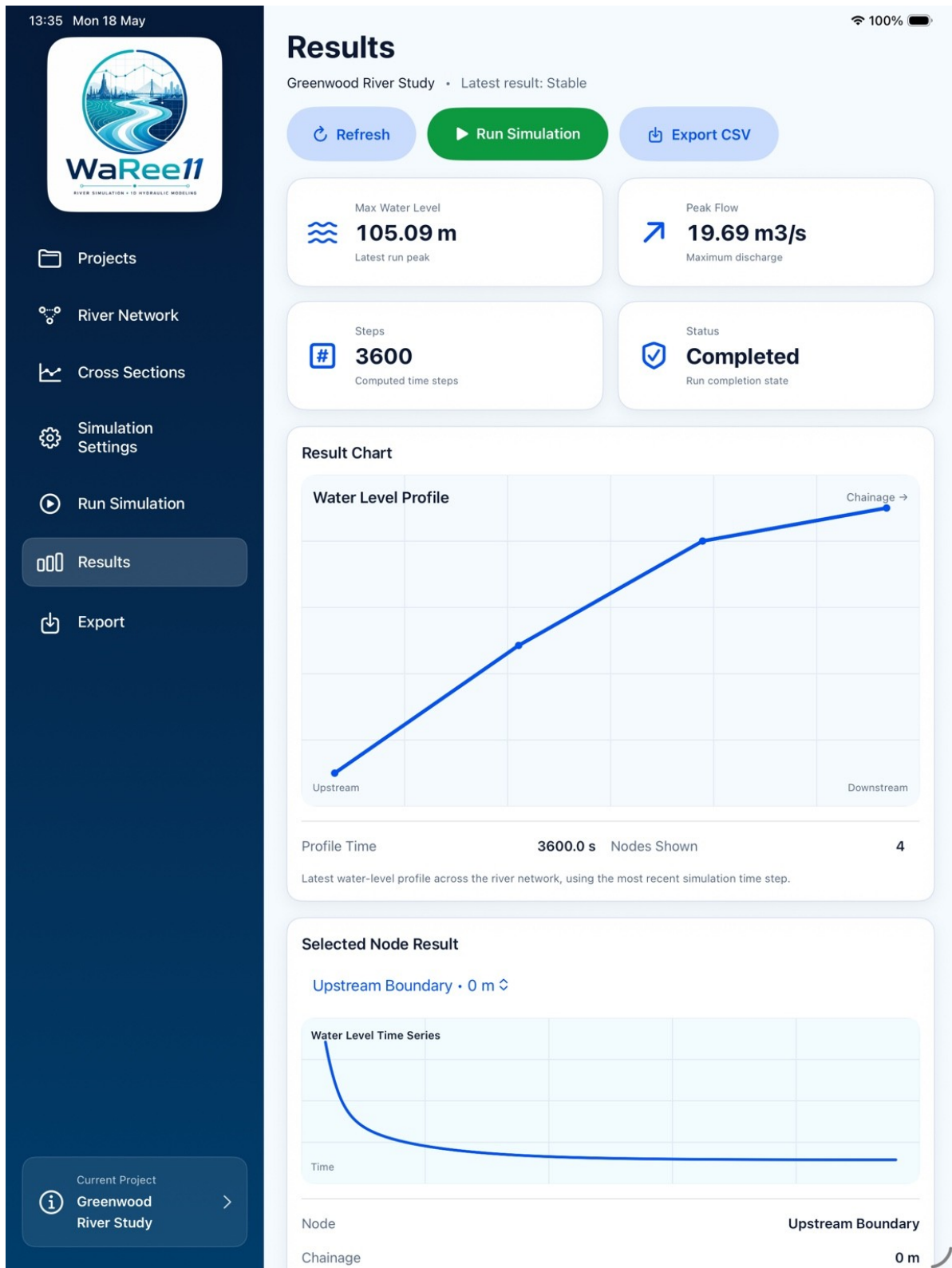


Figure 5. Results screen showing water level profile, peak flow, run status, and output readiness.

5. Technical Direction

WaRee11 is planned around a modern SwiftUI and MVVM architecture with repository-based data management, project-based storage, a modular solver engine, diagnostics, and export layers.

The hydraulic solver direction begins with a focused 1D river hydraulic workflow based on the Saint-Venant equation family. Development should progress through educational prototypes, explicit scheme comparisons such as MacCormack and Lax-Wendroff, finite-volume development, benchmark verification, and laboratory validation.

A central design principle is that numerical behavior should be visible. WaRee11 should help users understand time step selection, Courant number / CFL behavior, missing inputs, and whether a simulation result is suitable for interpretation.

6. GIS-Assisted Model Preparation

The future WaRee11 GIS editing layer should provide model-specific spatial editing for hydraulic schematization. The goal is not to build a full QGIS or ArcGIS replacement, but to support river nodes, branches, cross-section placement, roughness zones, attributes, GeoJSON/CSV interoperability, and hydraulic readiness checks inside the WaRee11 workflow.

7. Validation and Collaboration Roadmap

Code Verification → **Benchmark Testing** → **Inter-Model Comparison** → **Laboratory Validation**
→ **Field Validation**

WaRee11 is well suited to structured collaboration with university hydraulic laboratories, water engineering departments, research groups, irrigation and flood agencies, and professional engineering partners. A practical collaboration package could include a teaching pilot, a benchmark validation dataset, and a controlled hydraulic laboratory demonstration.

8. Development Roadmap

Phase	Focus	Key Output
Phase 1	App Store Prototype	Core workflow, diagnostics, scenario management, and export basics.
Phase 2	Hydraulic Engine Strengthening	Solver input builder, CFL calculator, scheme comparison, benchmark cases.
Phase 3	GIS Editing Layer	Map canvas, layer manager, node/branch editing, cross-section placement.
Phase 4	Calibration and Lab Validation	Benchmark library, lab datasets, inter-model comparison, validation reporting.
Phase 5	Education and Professional Platform	Teaching modules, partner pilots, documentation, training datasets, professional reporting.

9. Partner Ask

WaRee11 is seeking early collaboration and feedback in the following areas:

- Pilot course collaboration
- Benchmark validation datasets
- Hydraulic laboratory partners
- Technical feedback from river modeling experts
- Research collaboration on iPad-based hydraulic model preparation

WaRee11 can become a bridge between classroom learning, field observation, hydraulic model preparation, and professional river modeling practice.